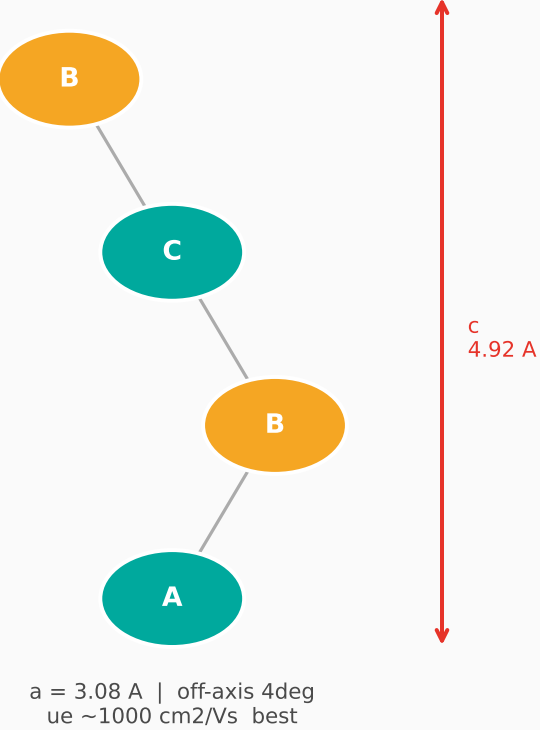
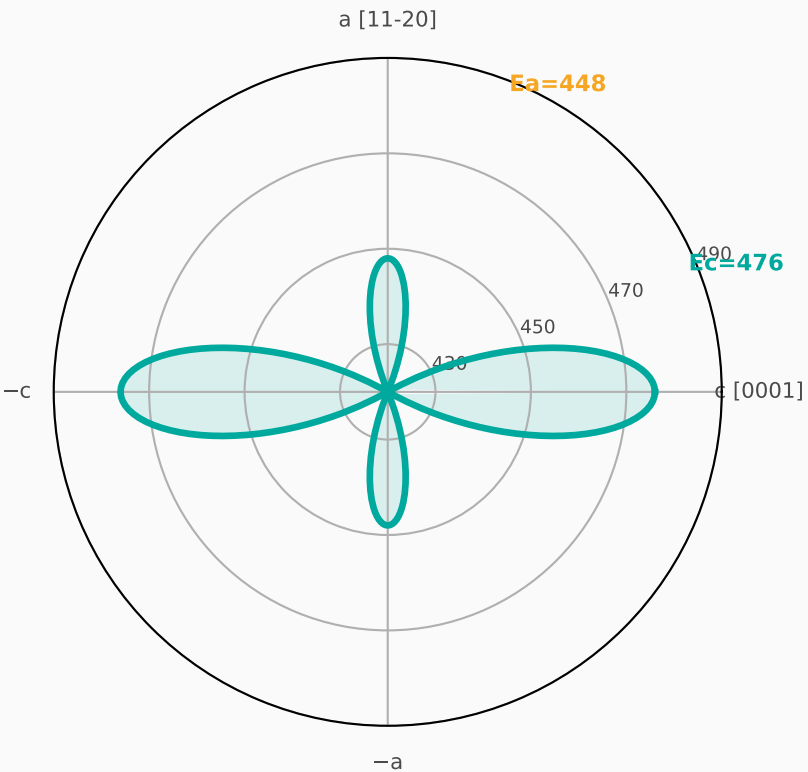


4H-SiC Crystal Anisotropy · Dicing Chipping Model

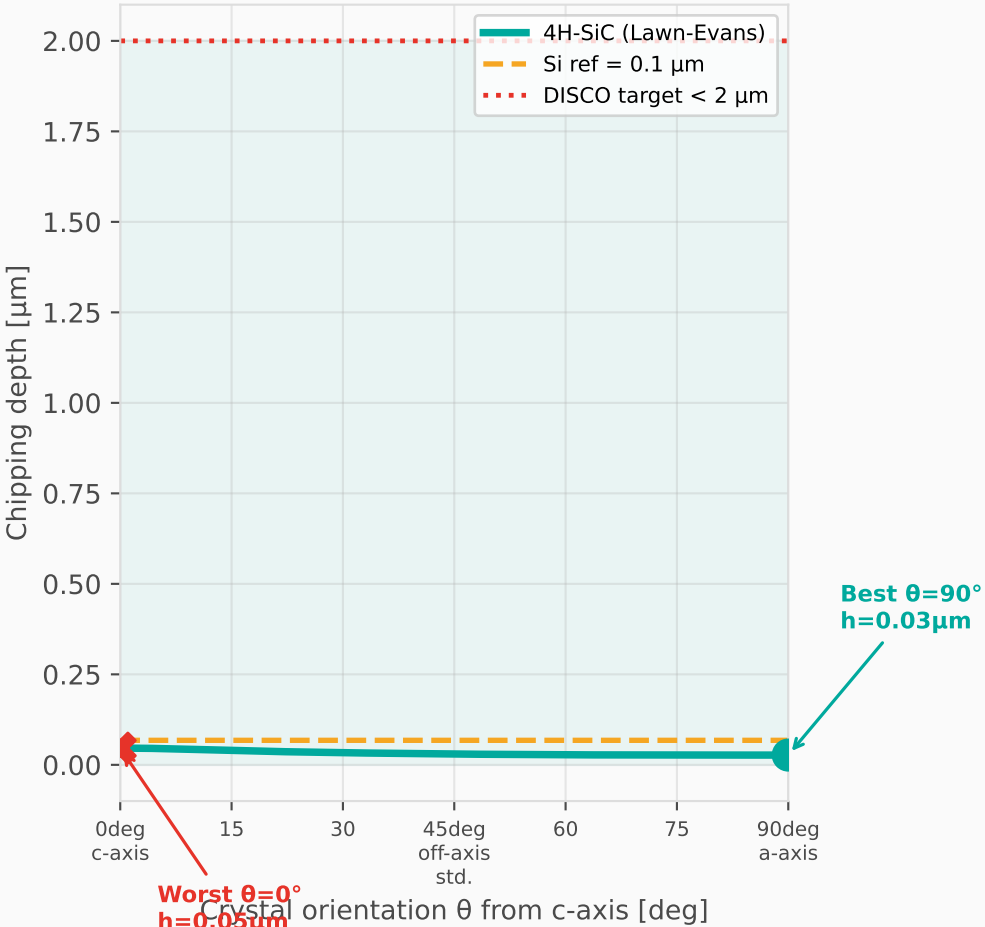
4H-SiC ABCB
(EV Power — DISCO main product)



Young's Modulus E(θ) [GPa]



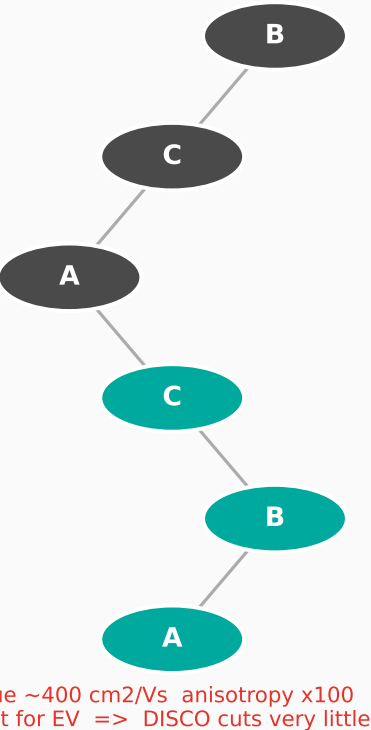
Chipping Depth h_{chip}(θ)
(depth=30μm, feed=1μm/rev, 25 krpm)



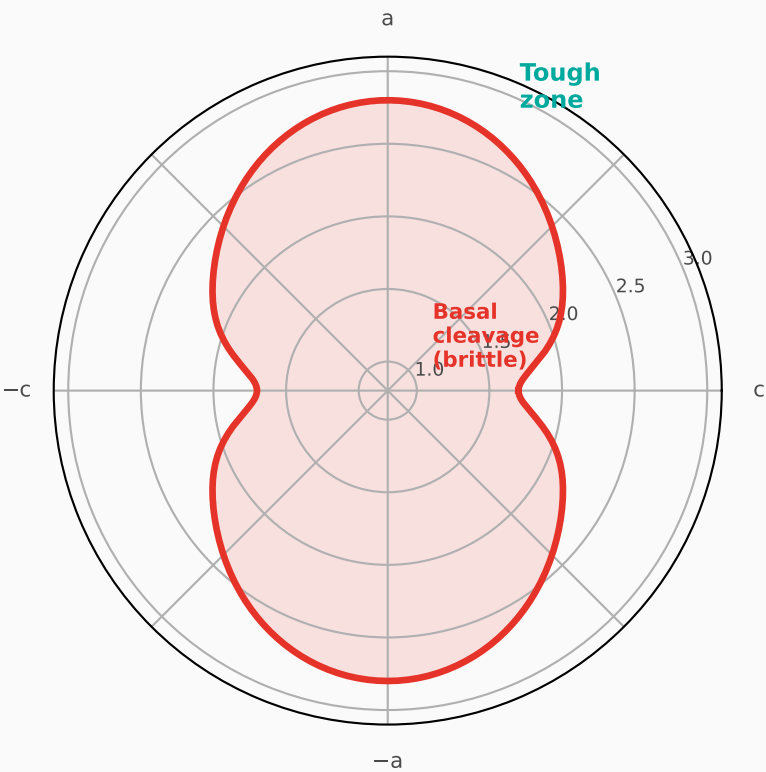
4H vs 6H vs Si: Key Properties

Property	4H-SiC	6H-SiC	Si
Bandgap [eV]	3.26	3.03	1.12
ue [cm ² /Vs]	~1000	~400	1400
u isotropy	Good	Poor x100	Isotropic
Ec [GPa]	476	~501	130
Ea [GPa]	448	~448	130
K _{IC} [MPa*sqtrm]	1.0 - 2.8	1.0 - 2.8	0.70
Hardness [GPa]	26 - 28	26 - 28	13
Etchrate (KOH)	Resistant	Resistant	Fast
EV power device	Best (main)	Not for EV	--
GaN-on-X sub	Good (growing)	Legacy	No
DISCO cut vol	Rapidly growing	Almost none	High
Blade wear rate	2x Si	2x Si	Baseline

6H-SiC ABCACB
(RF GaN substrate / niche)



Fracture Toughness K_{IC}(θ) [MPa√m]



Wafer Top View — Dicing Direction vs Chipping
(arc colour: green = low chipping, red = high)

